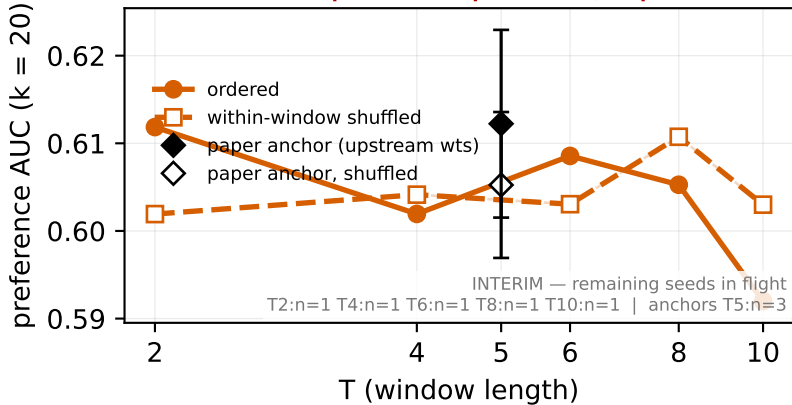


## G1 PORT-FIDELITY VERDICT PENDING — pre-G1 cells (pilot to 25k; no plateau at ~21k vs upstream 5.8k)



paper-faithful arm: agentic\_txc\_02\_v1t = the paper's own RLHF TXC architecture (MatryoshkaTXCDRContrastiveMultiscale) ported to this harness, on the paper stream (gemma\_2\_2b\_base\_l12\_phase7). Conclusions here are statements about the PAPER arm itself, not the plain-TXC modernization (that is the companion figure). Port fidelity is certified by the G1 verdict — see the coverage note for its state. KNOWN DEVIATIONS from the upstream recipe, disclosed not patched: (i) upstream clips gradients at 1.0; this harness applies NO gradient clipping. (ii) rows record precision='bf16' but training is fp32 throughout — the field is declarative only (no autocast/GradScaler). Everything else — const Adam 3e-4, no warmup, batch schedule, k\_win=100·T, step budget — follows upstream.